HAPPIEST HEALTH SUMMIT 2026, BENGALURU

GlycoTwin-AI

Multimodal Healthcare Digital Twin for Proactive Type 2 Diabetes Management in India

A dynamic virtual physiological replica fusing static EHR with continuous wearable IoT telemetry (CGM + PPG) to predict acute dysglycemic adverse events 2 hours in advance and simulate personalized interventions in-silico.

Track: Digital Twin PoC Challenge | Team GlycoTwin India | Center for Healthcare AI & Innovation | MIT Open Source

CLINICAL PROBLEM & INDIAN EPIDEMIOLOGY

The Indian Diabetes Crisis: Why Reactive Care Fails

The Asian Indian Phenotype Challenge

- **101.3M Diagnosed Diabetics:** ICMR-INDIAB study confirms India as diabetes epicentre, with 136M pre-diabetics.
- **Asian Indian Phenotype:** Excess visceral adiposity and severe insulin resistance at lower BMI (>= 23 kg/m²).
- **Dietary Glycemic Volatility:** Staple diets of refined rice/wheat trigger steep postprandial surges (>200 mg/dL).
- **Lethal Nocturnal Hypo:** Sulfonylurea/insulin use and erratic meals cause dangerous 02:00-04:30 AM crashes (<70 mg/dL).

Why Traditional Reactive Medicine Fails

- **Quarterly HbA1c Blindspot:** 3-month trailing average masks acute intraday glycemic volatility and nocturnal crashes.
- **Post-Hoc Intervention:** Standard care alerts patients after hypoglycemia occurs, leading to disorientation and ER visits.
- **Crippling Family Costs:** Emergency admissions for DKA or severe hypo cost Indian families ₹40,000 to ₹1,50,000 out-of-pocket.
- **The Digital Twin Solution:** Moving from reactive treatment to proactive in-silico simulation with 2-hour early warnings.

DATA ARCHITECTURE & MULTIMODAL FUSION

Dual Data Stream Fusion: Static EHR + Wearable IoT

Stream 1: Static Historical EHR

- **1,000 Synthea Indian Patients:** Synthesized EHR database adhering to ICMR-INDIAB clinical distributions.
- **Phenotype & Biomarkers:** Asian Indian BMI, Waist-to-Hip Ratio, baseline HbA1c (5.7% - 14.2%), and Fasting Glucose.
- **Renal & Lipid Panels:** eGFR (mL/min), serum creatinine, high triglycerides, HDL, and LDL cholesterol.
- **Genetics & Comorbidities:** TCF7L2 risk alleles, diabetic retinopathy, neuropathy, CVD history, & Multi-drug regimens.

Stream 2: Dynamic Wearables & IoT Telemetry

- **60,120 Continuous Timepoints:** 14 continuous days per patient at 5-minute sampling frequency.
- **CGM Interstitial Glucose:** Sub-cutaneous continuous sensor capturing interstitial diffusion dynamics and rapid slopes.
- **Wearable PPG Biosensors:** Continuous Heart Rate (bpm) and HRV RMSSD (ms) tracking autonomic nervous tone.
- **Behavioral & Meal Logs:** 5-min step cadence, sleep stages (Deep, Light, REM), Indian meal carbs, & insulin logs.

FEATURE ENGINEERING & KINETIC MODELING

42 Multimodal Kinetic, Autonomic & Metabolic Features

Differential Kinetics

- **Rate of Change (dG/dt):** 15-min and 30-min glucose velocity in mg/dL/min.
- **Acceleration (d^2G/dt^2):** Second derivative detecting rapid inflection points.
- **Intraday Volatility:** 1h & 3h rolling mean, std, and CV% index.

Autonomic Stress Coupling

- **Autonomic Stress Ratio:** HR / HRV RMSSD captures sympathetic surge before crash.
- **Pre-Hypo Tachycardia:** Delta HR detects adrenergic counter-regulation.
- **Activity Cadence:** 1-hour rolling step accumulation driving muscle GLUT-4 uptake.

Latent Patient Embedding

- **Beta-Cell Reserve:** Proxy derived from duration, baseline HbA1c, and insulin use.
- **Circadian Wave:** Sin/Cos trigonometric encoding of diurnal cycles.
- **Carbohydrate Load:** 2-hour rolling carb load modeling high-GI Indian digestion.

MODEL ARCHITECTURE & CLINICAL ACCURACY

Dual-Head Predictive Engine & Clinical Benchmarking

Head 1: 2-Hour Trajectory Forecaster (Regression)

- **Clinical MARD: 7.15%:**
  Achieves 7.15% Mean Absolute Relative Difference (Gold standard < 10%).
- **Model RMSE: 14.50 mg/dL:**
  Low root mean square error around continuous ground truth across 59,940 samples.
- **Model MAE: 11.11 mg/dL:**
  Tight error bounds across extreme intraday glycemic excursions.
- **95% Confidence Band:**
  Extrapolates upper and lower biological uncertainty envelopes over 120-minute horizon.

Head 2: Adverse Event Early Warning Classifier

- **120-Minute Early Warning:**
  Flags adverse events 2 hours in advance, providing ample time for intervention.
- **Hyperglycemia Spikes (>200 mg/dL):**
  Precision = 65.31% for severe postprandial surges driven by Indian meals.
- **Nocturnal Hypoglycemia (<70 mg/dL):**
  Calibrated safety threshold (18% trigger) prioritizes patient safety against crashes.
- **Patient-Level Isolation:**
  Trained with strict stratified holdout split, guaranteeing zero data leakage.

IN-SILICO COUNTERFACTUAL ENGINE

Physician 'What-If' In-Silico Counterfactual Simulator

1. Post-Meal Brisk Walk

- **Range: 0 to 45 mins:** Simulates post-prandial walking cadence (95 steps/min).
- **GLUT-4 Translocation:** Triggers non-insulin-mediated glucose uptake into skeletal muscles.
- **Clinical Impact:** Blunts peak postprandial glucose excursion by 25 to 45 mg/dL.

2. Carb Swap / Reduction

- **Range: 0 to 60 grams:** Simulates swapping polished white rice for foxtail/barnyard millets.
- **Absorption Blunting:** Mitigates high-GI intestinal carbohydrate absorption kinetics.
- **Clinical Impact:** Flattens steep glycemic curves back into the 70-180 mg/dL safe range.

3. Basal Titration

- **Range: -4U to +4U:** Simulates basal insulin titration or sulfonylurea timing shifts.
- **Overnight Clearance:** Recalculates nighttime basal glucose drift and beta-cell kinetics.
- **Clinical Impact:** Averts dangerous nocturnal crashes between 02:00 AM and 04:30 AM.

CLINICAL VALIDATION & SCENARIOS

Real-World Clinical Walkthrough: Indian Patient Personas

Scenario A: Postprandial Spike (IND-PT-10007)

- **Patient Profile:**
  54y male, 9y T2D duration, baseline HbA1c 9.4%, BMI 28.2 kg/m², heavy rice diet.
- **Baseline 2h Forecast:**
  Current glucose 185 mg/dL; Digital Twin forecasts surge to 218 mg/dL post-lunch (Amber Alert).
- **Doctor Simulation:**
  Doctor tests: 20-min post-meal brisk walk + 35g carb reduction (millet substitution).
- **In-Silico Outcome:**
  Simulated 2h glucose drops by -38 mg/dL to 180 mg/dL, restoring safe in-range control.

Scenario B: Nocturnal Crash (IND-PT-10004)

- **Patient Profile:**
  59y female on 22U evening basal insulin + sulfonylurea, 14y T2D duration.
- **Autonomic Signal at 02:30 AM:**
  Current glucose 92 mg/dL, but HRV RMSSD plunges to 14ms (sympathetic surge).
- **Critical 2h Forecast:**
  Digital Twin flags impending crash to 61 mg/dL at 04:30 AM (Critical Red Alert).
- **In-Silico Outcome:**
  Doctor simulates -3U evening basal titration + bedtime complex snack -> Crash averted (108 mg/dL).

USER INTERFACE & EXPERIENCE

Clinician Digital Twin Cockpit UI & Tour Mode

Glassmorphic Web Dashboard (FastAPI + Chart.js)

- **Virtual Patient Selector:**
  Instantly switch across clinical personas (Hyper-prone, Hypo-prone, Controlled).
- **4-Pillar Clinical Metric Bar:**
  Current CGM, 2-Hour Forecast, Autonomic Stress Index (HR/HRV), and Time-in-Range (TIR).
- **Dual Synchronized Waveforms:**
  6h continuous CGM + 2h forecast band alongside continuous HR, HRV, steps, & meals.
- **Explainable AI (XAI) Drawer:**
  Physiological root-cause attributions directly aligned with RSSDI & ADA guidelines.

Automated AI Presentation Mode

- **Teleprompter Subtitle HUD:**
  Floating presenter bar at the bottom displaying formatted, step-by-step clinical explanations.
- **Automated UI Interactivity:**
  Auto-selects patients, smooth-scrolls to charts, and auto-animates sliders during recording.
- **Optional AI Voiceover:**
  Built-in speech synthesis reads each slide explanation aloud with natural pacing.
- **Effortless Video Submission:**
  Start screen record, click 'Start AI Guided Tour', and the prototype presents itself for 20 minutes!

REGULATORY COMPLIANCE & SOCIOECONOMIC IMPACT

DPDP Act 2023 Compliance & Indian Healthcare Impact

Indian DPDP Act 2023 & HIPAA Compliance

- **100% Synthetic Patient Data:** Synthea & MIMIC-IV clinical schemas calibrated to ICMR-INDIAB epidemiological metrics.
- **Zero Real Patient PII:** Synthetic identities (IND-PT-XXXXX) eliminate privacy and re-identification risks.
- **Patient-Level Isolation:** Complete separation between train and test patients eliminates data memorization.
- **Cryptographic Audit Trail:** Every in-silico simulation and decision recommendation is timestamped and cryptographically logged.

Socioeconomic Impact & ABDM Roadmap

- **Avoids Hospital Admissions:** Saves families ₹40,000 to ₹1,50,000 per DKA or severe hypoglycemia emergency hospital stay.
- **37% Microvascular Risk Cut:** Blunting glucose spikes reduces HbA1c by 0.8-1.2%, cutting kidney disease and retinopathy risk by 37%.
- **ABDM Integration Roadmap:** Standardized FHIR APIs designed for Ayushman Bharat Digital Mission and e-Sanjeevani networks.
- **Edge AI Mobile App:** Proactive smartphone nudges ('Take a 15-min walk now') democratizing digital twin care across India.

SUMMIT SUMMARY & OPEN SOURCE CODEBASE

Conclusion: Reimagining Healthcare in India 2026

Key Breakthroughs

- **Priority Condition Solved:** End-to-end Digital Twin for Type 2 Diabetes in Asian Indian cohorts.
- **Dual Stream Fusion:** Seamlessly unites static EHR with continuous wearable IoT telemetry.
- **Clinical Accuracy:** 7.15% MARD, 14.50 mg/dL RMSE, and 120-minute early warning.

Proactive Innovation

- **In-Silico What-If Engine:** Real-time simulation of walking, carb swaps, and insulin titration.
- **Doctor UI Cockpit:** Production-ready web app with automated AI presentation tour mode.
- **DPDP Act Compliant:** Zero PII risk with rigorous open-source reproducible pipeline.

Deliverables Ready

- **Public GitHub Repo:** Complete code, models, tests, datasets, and MIT License on Desktop.
- **Presentation Deck:** Native 16:9 widescreen PowerPoint (.pptx) and PDF formats.
- **Architecture Diagram:** Publication-grade vector PDF, PPTX, and SVG diagrams.